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Assignment-2

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1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Hardware Issue	Typical Symptom
1. RAM failure	PC does not boot, blank screen, or repeated beep sounds
2. Hard disk/SSD failure	Slow system, files not opening, or OS not booting
3. Power supply (SMPS) failure	PC does not turn on or suddenly shuts down
4. CPU overheating	PC becomes very slow, freezes, or shuts down automatically
5. Graphics card failure	No display, screen artifacts, or display driver errors

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step	What to Check	What to Look For
1. Check power socket	Plug another device into the socket.	Make sure the socket is providing power.
2. Check charger	Inspect the charger, cable, and connector.	Look for cuts, loose wires, or damage.
3. Check charging port	Check the laptop charging port.	Look for dust, damage, or a loose port.
4. Check charger LED	If the adapter has an LED, check it.	LED should normally be ON when connected to power.
5. Try another compatible charger	Use a known-good charger with the correct voltage/current.	If it charges, the original charger may be faulty.
6. Remove battery (if removable)	Turn off the laptop and remove the battery. Then connect the charger.	If the laptop turns on, the battery may be faulty.
7. Check with a multimeter	Measure the charger's output voltage.	Compare the reading with the voltage printed on the charger.
8. Final check	If charger and battery seem fine, inspect the charging circuit/port.	A faulty charging port or motherboard charging circuit may need repair.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Test	Result
Charger type	Standard USB 5V charger
Multimeter setting	DC Voltage (V $\equiv$)
Expected voltage	4.75V – 5.25V
Measured voltage	5.02V
Result	☒ Within expected range

Conclusion: The measured output of **5.02V** is within the expected **4.75V–5.25V** range, so the charger’s voltage output is normal.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

ATX Power Supply 12V Rail Test

Steps:

1. Shut down the PC and **unplug the power supply from the wall.**
2. Disconnect the **ATX connector** from the motherboard.
3. Set the multimeter to **DC Voltage (V=)**.
4. Identify the **yellow wire (+12V)** and **black wire (Ground)**.
5. Connect the **black probe to the black wire** and the **red probe to the yellow wire.**
6. Read the voltage on the multimeter.

Example reading: 11.2V

Test	Result
Expected 12V rail	Around 12V
Measured voltage	11.2V
Below 11.5V?	Yes ✗
Possible issue	Faulty/weak PSU (SMPS)